

Covariant Form of Ohm's Law

D. Tong, Electromagnetism, Problem Sheet 3, Question 10.

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Electrodynamics

For a general 4-velocity, written as $u^\mu = \gamma(c, \mathbf{v})$, show that

$$F_{\mu\nu}u^\nu = \gamma \begin{pmatrix} -\mathbf{E} \cdot \mathbf{v}/c \\ \mathbf{E} + \mathbf{v} \times \mathbf{B} \end{pmatrix}. \quad (1)$$

In the rest-frame of a conducting medium, Ohm's law states that $\mathbf{J} = \sigma \mathbf{E}$, where σ is the conductivity and $\mathbf{J}$ is the 3-current. Assuming that σ is a Lorentz scalar, show that Ohm's law can be written covariantly as

$$J^\mu + \frac{1}{c^2}(J^\nu u_\nu)u^\mu = \sigma F^{\mu\nu}u_\nu, \quad (2)$$

where J^μ is the 4-current and u^μ is the (uniform) 4-velocity of the medium. If the medium moves with 3-velocity $\mathbf{v}$ in some inertial frame, show that the current in that frame is

$$\mathbf{J} = \rho \mathbf{v} + \sigma \gamma \left(\mathbf{E} + \mathbf{v} \times \mathbf{B} - \frac{1}{c^2}(\mathbf{v} \cdot \mathbf{E})\mathbf{v} \right), \quad (3)$$

where ρ is the charge density. Simplify this formula, given that the charge density vanishes in the rest-frame of the medium.

Solution It is easy to verify that

$$F_{\mu\nu}u^\nu = \gamma \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix} \begin{pmatrix} c \\ v_x \\ v_y \\ v_z \end{pmatrix} = \gamma \begin{pmatrix} -\mathbf{E} \cdot \mathbf{v}/c \\ \mathbf{E} + \mathbf{v} \times \mathbf{B} \end{pmatrix}. \quad (4)$$

In the rest-frame, $u^\mu = (c, 0, 0, 0)$. Then, consider

$$J^\mu + \frac{1}{c^2}(J^\nu u_\nu)u^\mu = \begin{pmatrix} \rho c \\ \mathbf{J} \end{pmatrix} + \frac{1}{c^2}(-\rho c \quad \mathbf{J}) \cdot \begin{pmatrix} c \\ 0 \end{pmatrix} \begin{pmatrix} c \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ \mathbf{J} \end{pmatrix}, \quad (5)$$

and

$$\sigma F_{\mu\nu}u^\nu = \begin{pmatrix} 0 \\ \sigma \mathbf{E} \end{pmatrix} \quad (6)$$

Thus, we have verified that

$$J^\mu + \frac{1}{c^2}(J^\nu u_\nu)u^\mu = \sigma F^{\mu\nu}u_\nu \quad (7)$$

is indeed the covariant representation of Ohm's law.

For a moving medium, the equation gives

$$\begin{pmatrix} \rho c + \gamma^2(-\rho c^2 + \mathbf{J} \cdot \mathbf{v}) \\ \mathbf{J} + \gamma^2(-\rho c^2 + \mathbf{J} \cdot \mathbf{v})\mathbf{v}/c \end{pmatrix} = \gamma \sigma \begin{pmatrix} \mathbf{E} \cdot \mathbf{v}/c \\ \mathbf{E} + \mathbf{v} \times \mathbf{B} \end{pmatrix}. \quad (8)$$

Combining the two equations, we find

$$\mathbf{J} = \rho \mathbf{v} + \sigma \gamma \left(\mathbf{E} + \mathbf{v} \times \mathbf{B} - \frac{1}{c^2}(\mathbf{v} \cdot \mathbf{E})\mathbf{v} \right). \quad (9)$$

Finally, given $\rho_0 = 0$, then $J^\mu u_\mu = -\rho_0 c^2$. This implies

$$-\rho c^2 + \mathbf{J} \cdot \mathbf{v} = 0, \quad (10)$$

Taking the dot product of $\mathbf{J}$ with $\mathbf{v}$ gives

$$\rho v^2 + \sigma \gamma \left(\mathbf{v} \cdot \mathbf{E} - \frac{v^2}{c^2} \mathbf{v} \cdot \mathbf{E} \right) = \rho c^2. \quad (11)$$

Rearranging, we find

$$\rho = \frac{\sigma \gamma}{c^2} \mathbf{v} \cdot \mathbf{E}. \quad (12)$$

Substituting back, the current becomes

$$\mathbf{J} = \sigma \gamma (\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (13)$$